

Rahul Santosh Shinde

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SUMMARY

- **Software Engineer** with around **4 years of experience** in full-stack development, specializing in **cloud-based solutions**, **data integration**, and **healthcare applications**.
- **Skilled** in designing, developing, and deploying **RESTful APIs**, **microservices**, and **real-time data systems** for healthcare platforms, ensuring seamless interoperability and compliance with **HIPAA**, **HL7**, and **FHIR** standards.
- **Expertise** in leveraging **AWS**, **Azure**, **Google Cloud**, and modern containerization tools like **Docker** and **Kubernetes** to optimize **scalability**, **performance**, and **cloud infrastructure**.
- **Engineered** secure and scalable applications by integrating **data security** measures such as **OAuth 2.0**, **JWT**, and **encryption** protocols, safeguarding sensitive patient data.
- **Developed** innovative **AI** and **machine learning** models, improving predictive analytics, and **enabled real-time video communication** using **WebRTC** for telemedicine applications.
- **Collaborated** in **Agile** teams to deliver impactful software solutions that enhance user experience and performance, consistently exceeding business goals.
- **Reduced integration errors** by 30% through process optimization, enhancing data flow and system reliability in healthcare data systems.
- **Implemented** automated workflows using **CI/CD** practices, streamlining development cycles and boosting efficiency.
- **Built decentralized applications** using **Ethereum smart contracts**, ensuring secure data handling and user interactions in blockchain-based projects.
- **Driven** by a passion for innovation, continuously staying at the forefront of emerging technologies to deliver solutions that improve healthcare, data security, and system performance.

SKILLS

- **Languages:** Python, JavaScript (ES6+), Java, TypeScript, C++, Kotlin, HTML5, CSS3, SQL, Shell Scripting.
- **Frontend Frameworks:** React, Angular, Vue.js, Bootstrap, Material UI.
- **Backend Frameworks:** Flask, Django, Express.js, Spring Boot, Node.js, NestJS.
- **Databases:** MySQL, PostgreSQL, MongoDB, SQLite, Firebase, Oracle SQL.
- **Cloud & DevOps:** AWS, Azure, Google Cloud, Docker, Kubernetes, Terraform, Jenkins, GitLab CI, CircleCI.
- **Version Control & Collaboration:** Git, GitHub, GitLab, Jira, Confluence, Trello.
- **Methodologies:** Agile, Waterfall, Scrum
- **Healthcare Tools & Technologies:** HIPAA, EMR, HER
- **Data Science:** Supervised & Unsupervised Learning, Predictive Learning, Neural Networks, Model Evaluation and Tuning, EDA, Tableau, Power BI, ETL Development & Optimization, Data Mining & Cleansing
- **Other Technologies:** GraphQL, WebSockets, WebRTC, RESTful APIs, Microservices, Firebase, RabbitMQ, Redis.

EXPERIENCE

CVS Health

August 2024 – Present

Software Engineer

- **Developed and implemented RESTful APIs** using **Django** to integrate **EHR** systems with third-party healthcare applications, ensuring compliance with **HL7** and **FHIR** standards.
- **Designed cloud-based architecture** on **AWS**, utilizing **S3** for secure data storage and **EC2** instances to process high-volume healthcare data transactions efficiently.
- **Implemented data security measures** including **OAuth 2.0**, **JWT tokens**, and **data encryption**, ensuring HIPAA compliance and safeguarding patient information.
- **Streamlined data flow** between healthcare systems, reducing integration errors by 30% and improving system performance with real-time data transformation using **AWS Lambda**.
- **Collaborated with cross-functional teams** to define technical requirements and ensure the platform met industry standards, enhancing interoperability and reducing administrative delays.

- **Spearheaded** the creation of a **HIPAA-compliant** patient management system, integrating appointment scheduling and medical record tracking features.
- **Engineered RESTful APIs** using **Spring Boot** to interface with **EHR** and **HL7** systems, facilitating seamless data exchange.
- **Designed** a responsive **frontend** with **React**, enhancing user experience for both patients and healthcare providers.
- **Secured** sensitive patient data by adhering to **HIPAA** guidelines and **implementing** encryption protocols.
- **Optimized** cloud infrastructure with **AWS**, enhancing data storage scalability and enabling effective disaster recovery.
- **Developed** a medication tracking platform with **Django** and **Twilio**, providing **real-time reminders** to patients.
- **Boosted** system performance by **leveraging Redis** caching for rapid access to medication adherence data.
- **Implemented WebRTC** for **real-time video consultations**, allowing patients and doctors to interact remotely.
- **Architected** the **backend** with **MongoDB**, managing patient records and appointment histories efficiently for telehealth services.
- **Automated** the deployment process using **Jenkins**, **Docker**, and **CI/CD** practices, reducing development cycles and improving release management.

PROJECTS

Lead Developer, Video GPT: Conversational Video Q&A with RAG and LLMs

- Engineered a Gradio web app for conversational video interaction, integrating Whisper for transcription and GPT-3.5-turbo for dialogue.
- Utilized Langchain’s Retrieval-Augmented Generation (RAG) to enhance query accuracy and search functionality.
- Deployed the app using Flask for efficient backend support and dynamic real-time video processing.

Full Stack Developer, Campus Flow: USC Door Data Revolution

- Spearheaded the development of a React-based web app for automating security data collection from 75,000 access points across USC campus.
- Utilized MongoDB for backend storage and integrated OpenCV for image analysis, optimizing data collection and system performance.

Lead Programmer, Drushti: AI Navigator for the Visually Impaired

- Led the creation of an AI-based indoor navigation system for the visually impaired, achieving a 12% error rate in obstacle detection.
- Developed a robust ensemble model for object detection with increased accuracy (67%) using YOLOv3, GoogleNet, and AlexNet.

Coding Head, Ether Vote: Decentralized Voting Blockchain Application

- Developed a decentralized voting app using Ethereum smart contracts with Solidity and Web3.js, ensuring secure data handling and user interactions.
- Achieved 99% contract accuracy with MythX, Truffle, and Ganache security verifications.

EDUCATION

Master of Science in Computer Science University of Southern California - Los Angeles, CA Relevant Coursework: Algorithms, Web Technologies, AI, Database Systems, Robotics.	AUG 2022 to MAY 2024
Bachelor of Engineering in Computer Engineering Fr. Conceicao Rodrigues College of Engineering - Mumbai, India Relevant Coursework: Big Data, Distributed Systems, Cloud Computing, Data Structures, Operating Systems.	AUG 2018 to MAY 2022